

Hierarchically-Adaptive Global-Local Grouping for Multi-Agent Reinforcement Learning in Urban Mobility Resource Rebalancing

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Abstract

We present HAGAS, a multi-agent reinforcement learning (MARL) framework that enables *hierarchical*, *adaptive*, and *scalable* policy sharing for city-scale mobility resource allocation. HAGAS adopts a global-local hierarchical structure across a trunk network and heads — that is, a shared backbone trunk network learns the global, city-level mobility patterns, while the light-weight policy heads enable local, group-level specialization — and performs trunk-head co-learning. To adaptively decide when and how to share the policy heads, HAGAS transforms and learns each agent’s experience based on its trajectory embeddings — i.e., a vector representation of its recent state-action-reward evolution — and performs drift-aware agent grouping that concerns adaptive split-merge operations. Specifically, regions with similar trajectory embeddings therefore share a policy head, while regions with diverging trajectories are split to retain local specialization. Furthermore, HAGAS establishes learnable identity embeddings to capture intra-group differences instead of direct parameter sharing. Taking urban micro-mobility sharing as instances, we have performed extensive experiments based on datasets of station-based bike and dockless electric scooter sharing services from New York City (NYC) and City of Austin that involve nearly 1.5M trips in total. We have demonstrated that thanks to our hierarchically-adaptive global-local grouping, HAGAS consistently improves service availability and overall allocation quality compared with state-of-the-art approaches. **Project Website:** github.com/FarshidNooshi/HAGAS-SIGSPATIAL2026.

Keywords

Multi-agent reinforcement learning, urban mobility resource rebalancing, hierarchically-adaptive grouping

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1 Introduction

Urban mobility resources, such as shared bikes and e-scooters, are crucial for spatio-temporal connectivities within the spacious

metropolitan areas. The service quality of operating these resources hinges on adaptive, timely, and effective rebalancing of the demand (e.g., bike pick-ups and trips) and the supply (e.g., available bikes for pick-ups), in response to the complex city environments (e.g., changed traffic and weather conditions) and varying mobility needs (e.g., regular commutes and occasional touring). Such rebalancing operations — i.e., relocating urban mobility resources from the over-served locations to the under-served ones (often via trucks) — often need to be through proactive, coordinated actions across multiple regions in the city [2, 3, 6, 8].

Recent advances in reinforcement learning (RL) have paved a new paradigm for urban mobility resource allocation under the framework of the Markov decision process (MDP). Via learning the dynamic interactions (redistribution of mobility resources) of the agent (e.g., city-wide mobility resource coordinator) with the urban mobility environments, RL-based approaches aim to adapt and actuate the urban mobility resource allocation toward the mobility needs. Among the RL approaches explored, the multi-agent reinforcement learning (MARL) has been recently explored due to its distributed coordination capabilities particularly within spacious urban environments. By accounting for multiple coordinators (e.g., bike and e-scooter sharing re-locators) as agents, MARL can serve as the resource distribution engine, observing the dynamic mobility resources and environment (states), and dynamically allocate the mobility resources (actions) through strategically coordinating these agents’ local behaviors — based on redistribution policies (e.g., when and where the available mobility resources should be re-allocated) within and across the city regions.

Despite the prior results, one central design question arises in regard to the adaptivity and scalability of coordinating multiple city regions — that is, how these MARL agents can *share* the redistribution policies for adaptive and scalable training, since these policies are often heavily parameterized (in a policy network) and consequently computationally expensive to learn in complex urban environments. Two important challenges remain for MARL.

Challenge A — Global-Local Adaptivity to Spatio-Temporal Demand-Supply Drifts. In order to distribute redistribution policy of the urban mobility resources across agents, one may consider *grouping* the agents that share the similar trajectories — that is, the histories of state-action-reward evolution — and respectively instantiate a set of policy networks for each agent group. For instance, prior work has explored selective sharing via static groupings [1], dynamic regrouping (e.g., periodically re-clustering of the agents) [7], and mask-based partial sharing [5]. These grouping methods, however, may not provide a self-adaptive, principled mechanism regarding evolving group number and their varying



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sizes. We note that the urban mobility patterns can change significantly due to local events (e.g., street congestion) and global trends (e.g., time of a day, day of a week). Given the subsequent demand-supply drifts, these methods either render the redistribution policies obsolete locally, or fail to react proactively, leading to stale parameter sharing or unstable learning outcomes globally across the city regions.

Challenge B – Global-Local Scalability in Heterogeneous Urban Mobility Resource Allocation. One is well aware that learning the urban mobility allocation policy for every agent is often not feasible given the spacious metropolitan area. This is mainly because of the facts that the highly heterogeneous urban mobility patterns often vary across close and distant regions, and change over short term and long horizon, making it hard to share the policy that actually fits. On the other hand, we also note that conventional sharing of urban mobility resource redistribution policy may not necessarily capture heterogeneous, often locality-specific roles of different agents. The performance of MARL may hence deteriorate given the large-scale urban mobility patterns. There arises a crucial need for a *scalable* decomposition in sharing policies across agent groups, such that an adequate but not excessive number of urban mobility resource redistribution policies is maintained as needed.

To overcome these two research challenges, we propose **HAGAS**, an MARL framework with **H**ierarchically **A**daptive **G**rouping and **A**gent **S**pecialization, for dynamic urban mobility resource allocation, with three major contributions:

(a) Spatio-Temporal Drift-Aware Agent Grouping for Global-Local Adaptivity: Toward global-local adaptivity, we have introduced an adaptive split-merge mechanism that updates the mobility resource redistribution policy sharing structure based on each agent’s trajectory embeddings, i.e., its compact vector representation of the state-action-reward evolution relevant to a city region. When the agent’s behaviors diverge or stabilize in response to adaptability under drifting demand and supply patterns, HAGAS captures the mutual similarities and performs adaptive splitting and merging operations of the agent groups that represent different regions of the city. This way, HAGAS gains the global-local adaptivity toward the drifts in the spatio-temporal mobility patterns.

(b) Scalability via Global-Local Hierarchy of Trunk-Head Co-Learning: Toward global-local scalability, we have designed within HAGAS a shared trunk network as well as multiple policy heads, and enabled a global-local hierarchy across the trunk network and heads. HAGAS performs trunk-head co-learning – that is, a shared backbone trunk learns the global, city-level mobility patterns, while the light-weight policy heads enable local, group-level specialization. We have designed the learnable identity embeddings that allow the agents (representing the different regions) inside the same agent group to retain consistent, local specialization. Via our proposed trunk-head co-learning, HAGAS achieves global-local scalability in allocating the urban mobility resources.

(c) Extensive Real-World Experimental Studies: Taking the bike sharing and dockless e-scooter sharing as two instances, we have conducted extensive real-world urban mobility resource data analytics and experimental studies over nearly 1.5 million micromobility trips in total from New York City (NYC) and the City of Austin, Texas, and corroborated the effectiveness, adaptivity, scalability, and efficiency of HAGAS.

2 Overview of HAGAS Framework

• **Spatial and Temporal Discretization.** We take into account, and discretize a target city area into a total of K different regions that are indexed by $\mathcal{K} = \{1, \dots, K\}$. We take into account the existing census tracts as our region discretization. A set of N agents serving as the mobility resource re-allocators (e.g., a fleet of trucks or coordinators for bike rebalancing) is denoted by $\mathcal{A} = \{\mathcal{A}_1, \dots, \mathcal{A}_N\}$ ($N = K$ in our current studies), where each agent i coordinates the mobility resource inside a region at each time interval t . Here we discretize the time horizon into time intervals (e.g., days in our current studies) for the MARL formulation. In our MARL setting, an episode, i.e., a finite interaction cycle, spans over a total of T consecutive days (say, $T=31$ for a month).

• **Neighborhood Graph.** In order to structure mobility resource allocation across adjacent regions for our MARL, we form an undirected adjacency graph $G = (\mathcal{K}, \mathcal{E})$, where an edge $(k, k') \in \mathcal{E}$ if two regions are geographically adjacent (sharing boundaries). The neighbors of a region k are therefore formally given by a set $\mathcal{N}(k)$, i.e., $\mathcal{N}(k) = \{k' \in \mathcal{K} : (k, k') \in \mathcal{E}\}$. We empirically cap the maximum neighborhood size of a region to a constant M (e.g., 32 for New York City and 16 for the City of Austin), and use masking and padding for regions with fewer neighbors. Via the neighborhood graph, we therefore formulate our urban mobility resource allocation into a finite-horizon multi-agent Markov game [4].

• **Actions & States.** We form the actions and states in order for HAGAS to interact with, and capture and learn the urban mobility patterns and the corresponding resource allocation policy.

Specifically, for each time interval t , let d_t^k and o_t^k respectively be the numbers of mobility resource pick-ups and drop-offs at a region indexed by $k \in \mathcal{K}$ at the time interval t . For each agent i at time interval t , we find the sizes of mobility resources (e.g., bikes and e-scooters) that will relocate from the region k to one of its geographically adjacent neighbors ($\mathcal{N}(k)$) within the neighborhood graph as $\mathbf{a}_t^k = [a_t^{(k,k')}]_{k' \in \mathcal{N}(k)} \in \mathbb{R}^{|\mathcal{N}(k)|}$. Here $a_t^{(k,k')}$, if nonnegative, represents the outflow of mobility resources from region k to its neighbors, and the inflow otherwise.

HAGAS finds and updates the availability of mobility resources for each region as $b_{t+1}^k = \max \left\{ b_t^k + \sum_{k' \in \mathcal{N}(k)} a_t^{(k',k)} - d_t^k + o_t^k, 0 \right\}$, where $\mathcal{N}(k)$ represents the adjacent regions of a region k .

Then, for each time interval t , we form a global state $\mathbf{s}_t$ that consists of region inventory, demand history, temporal features, and geospatial context. In our implementation, each agent observes a *local* feature vector instead of a single concatenated city-wide vector. For a region k at time interval t , the observed state is therefore given by $\mathbf{s}_t^k = [b_t^k, \mu_t^k, v_t^k, \sin \frac{2\pi h_t}{24}, \cos \frac{2\pi h_t}{24}, \sin \frac{2\pi d_t}{7}, \cos \frac{2\pi d_t}{7}, C^k]$, which has a total of 18 dimensions ($C^k \in \mathbb{R}^{N_C}$ and $N_C = 11$).

• **Reward Function:** Given a coordinated relocation decision $\mathbf{A}_t = \{\mathbf{a}_1^t, \mathbf{a}_2^t, \dots, \mathbf{a}_N^t\}$ from our N agents, the availability of mobility resources (inventory) in a region k (before serving the actual pick-up requests) at time interval t is formally given by $\tilde{b}_t^k = b_t^k + \sum_{k' \in \mathcal{N}(k)} a_t^{(k',k)}$, where $a_t^{(k',k)}$ is the signed net inflow contributed by an agent at region k' to another coordinating k . The fulfilled demand is then given by $S_t^k = \min\{d_t^k, \tilde{b}_t^k\}$, as the actual pick-ups should not exceed the availability of mobility resources at the agent k . The unfulfilled demand is then calculated

by $U_t^k = d_t^k - s_t^k = \max\{d_t^k - \tilde{b}_t^k, 0\}$. Based on above, we define the reward function for each agent k at time interval t as $r_t^k = \lambda(1 - \frac{U_t^k}{d_t^k + \varepsilon}) - \alpha \cdot \frac{U_t^k}{d_t^k + \varepsilon} - \beta \cdot \frac{\|a_t^k\|_1}{M_{\text{load}}}$. Here a small constant ε numerically prevents division by zero when $d_t^i = 0$. The three components (with parameters $\lambda, \alpha, \beta > 0$) inside the reward function r_t^k respectively measure (1) high service ratio ($1 - \frac{U_t^k}{d_t^k + \varepsilon}$): which aims to capture the fulfillment across the agents; (2) low under-served demand ($\frac{U_t^k}{d_t^k + \varepsilon}$): which indicates the under-served agents over the learning process; (3) low cost of relocation ($\frac{\|a_t^k\|_1}{M_{\text{load}}}$): This is proportional to the total size of relocated mobility resources capped by maximum relocation load $M_{\text{load}} > 0$ in order to regularize the relocation costs.

As formulated as a finite-horizon multi-agent Markov game, we present the objective of HAGAS as follows. For a horizon of a total of T time intervals, and a discount factor $\gamma \in (0, 1]$, HAGAS aims to find the MARL parameters θ that maximize the objective function of expected cumulative rewards, i.e., $J(\theta) = \mathbb{E}[\frac{1}{N} \sum_{k=1}^N \sum_{t=0}^{T-1} \gamma^t r_t^k]$.

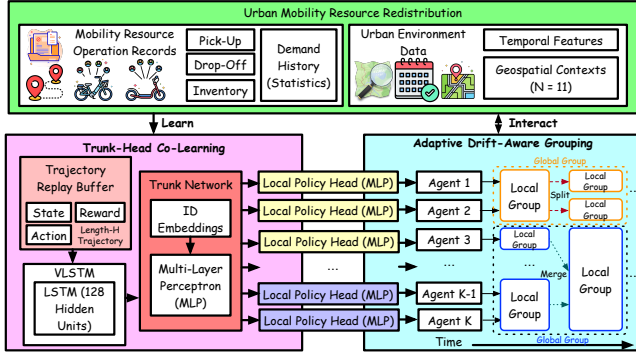


Figure 1: Overview of HAGAS.

3 Hierarchical Parameter Sharing

We present the overview of HAGAS framework in Fig. 1. In each time step, the agents observe the states and performs the actions of relocating urban mobility resources (e.g., bikes, e-scooters).

We present the detailed trunk-head co-learning designs as follows. Specifically, at each time interval t , a region (agent) k is assigned to one global group, which is indexed by $c_t \in \{1, 2, 3, \dots, C_{\text{max}}\}$ where C_{max} caps the total number of global groups. All agents in the global group c then share the parameters of the same trunk network T_{ψ_c} with parameters ψ_c : $\mathbf{h}_t^c = \text{Trunk}_{\psi_c}(s_t) \in \mathbb{R}^{d_h}$, which captures the hidden states via the d_h -dimensional embeddings $\mathbf{h}_t^c$. Within each global group c_t , the regions (agents) are further partitioned into local groups. Each local group, indexed by ℓ_t , operates a light-weight policy head $\pi_{\phi_{c_t, \ell_t}}$. Here, $\ell_t \in \{1, \dots, G_t^{c_t}\}$, $\sum_{c=1}^{C_{\text{max}}} G_t^{(c)} \leq G_{\text{max}}$, where G_{max} caps the total number of heads across all trunks.

To preserve local specialization during the parameter sharing, each region (agent) is differentiated by the learnable identity (ID) embeddings $\mathbf{E}_k \in \mathbb{R}^{d_E}$ (with $d_E=32$ in our experiments) of an agent coordinating a region. In our implementation, we use the simple embedding layer to encode the index of the region (agent), i.e.,

$\mathbf{E}_k = \text{Embed}(k)$, which will be learned dynamically throughout the MARL process.

We then find the shared parameters from the global trunk network with the agent k as $\mathbf{z}_t^k = [\mathbf{h}_t^c; \mathbf{E}_k]$. In each local group indexed by ℓ_t , we form the local policy head network in order to coordinate the mobility resource allocation. The action of an agent k at time interval t is therefore given by $\mathbf{a}_t^k \sim \text{Head}_{\phi_{c_t, \ell_t}}(\mathbf{z}_t^k)$. Here the action space is formed by a Gaussian actor network which outputs parameters (mean and standard deviation) of a Gaussian distribution to sample actions. In our implementation, we adopt the multi-layer perceptron (MLP) networks for both the global trunk network and the local policy head. We adopt 640 units for the hidden layer and 160 units for the output layer within the trunk network $\text{Trunk}(\cdot)$ — that is, $\mathbf{h}_t^c = \tanh(W_{\text{trunk}} s_t + b_{\text{trunk}})$, and multiple hidden units (160 for NYC and 192 for Austin) and the output dimension equivalent to $|\mathcal{N}(k)|$ (the action space for each agent k) for $\text{Head}_{\phi_{c_t, \ell_t}}(\cdot)$.

4 Adaptive Drift-Aware Grouping

We represent each region's recent behavior using a trajectory embedding. Specifically, for region k , we define a length- H trajectory ending by t as $\tau_{t-(H-1):t}^k = \{(s_{t-(H-1)}, \mathbf{a}_{t-(H-1)}^k, r_{t-(H-1)}^k), \dots, (s_t, \mathbf{a}_t^k, r_t^k)\}$, where s_t^k represents the normalized state vector (fed for proximal policy optimization), $\mathbf{a}_t^k$ is the executed action of agent k , and r_t^k denotes the reward received by region k at time t . When fewer than H steps are available, the trajectory is left-padded with zeros. We encode $\tau_{t-(H-1):t}^k$ using a very long short-term memory (VLSTM) network with parameters ω to obtain the trajectory embedding of agent k at time interval t .

We stack the trajectories of all agents belonging to a local group ℓ as $\Phi_t^\ell = [\tau_{t-(H-1):t}^k]_{k \in \mathcal{G}_t^\ell} \in \mathbb{R}^{|\mathcal{G}_t^\ell| \times H \times F}$, where F denotes the per-step feature dimension of the trajectory representation. The output embeddings for all agents in local group ℓ are then obtained as $\Omega_t^\ell = \text{VLSTM}_\omega(\Phi_t^\ell)$, where the final hidden state of the VLSTM is projected to its mean representation and used as the trajectory embedding for subsequent split-merge regrouping. Similar to the action space, we also sample the distribution from the output of VLSTM. Specifically, along the entire H -step trajectory, we find the mean embedding and the standard deviation of embeddings from the VLSTM as (μ_t^k, σ_t^k) to characterize the behaviors of an agent (region) k . Here we use this variational latent distribution, instead of solely using the vectorized embeddings (representing a point in the high-dimensional space), as such a distribution helps yield a principled drift measurement regarding the divergence. In addition, for each local group, we calculate the average embeddings as $\mu_t^{(\ell_t)} = \frac{1}{|\mathcal{G}_t^{(\ell_t)}|} \sum_{k \in \mathcal{G}_t^{(\ell_t)}} \Omega_t^k$, and find the standard deviations of the embeddings as $\sigma_t^{(I)}$ (empirically clamped by $[1e-3, 2.0]$). Based on the above, we further form the variational latent distribution in terms of the normal distribution for both the local group l as well as each agent k , i.e., $\mathcal{Q}_t^\ell = \mathcal{N}(\mu_t^{(\ell_t)}, \text{diag}((\sigma_t^{(\ell_t)})^2))$, and $\mathcal{P}_t^k = \mathcal{N}(\mu_t^k, \text{diag}((\sigma_t^k)^2))$.

Based on both distributions, we calculate and find the Kullback-Leibler (KL) divergence to capture the drift signals within the agent's behaviors and trajectories compared with the collective behaviors inside the local group. We specifically find the average symmetrized KL divergence ($\text{KL}(\cdot \| \cdot)$) within the group (namely the intra-group divergence).

Table 1: Overall performance & ablation of HAGAS.

Dataset	Methods	MRA (%) ↑	MPV ↓	MDPV (km) ↓	A-D Score ↑
Austin Baselines	Independent PPO (No-Share)	41.88	1.42	3.83	0.785
	Shared PPO (All-Share)	42.69	1.88	4.89	0.793
	Kaleidoscope	44.90	1.06	2.56	0.856
	Static Grouping	43.23	5.82	15.16	0.785
	Dynamic Sharing	45.08	5.29	12.37	0.820
Austin Ablations	HAGAS _{w/o} ID	36.00	5.15	13.53	0.656
	HAGAS _{w/o} SM	43.25	5.22	13.16	0.786
	HAGAS _{w/o} HG	38.75	4.35	10.30	0.708
	HAGAS _{w/o} ARP	39.25	5.28	13.54	0.714
	HAGAS _{w/o} VLSTM	40.50	3.27	7.58	0.744
	HAGAS	52.91	5.64	13.31	0.960
NYC Baselines	Independent PPO (No-Share)	45.17	1.68	1.29	0.924
	Shared PPO (All-Share)	43.02	1.75	1.50	0.875
	Kaleidoscope	43.04	1.80	1.54	0.875
	Static Grouping	44.71	6.47	4.98	0.878
	Dynamic Sharing	43.71	5.96	4.54	0.860
NYC Ablations	HAGAS _{w/o} ID	41.50	5.34	4.14	0.819
	HAGAS _{w/o} SM	43.75	6.36	4.89	0.860
	HAGAS _{w/o} HG	45.50	6.32	4.95	0.893
	HAGAS _{w/o} ARP	43.75	6.13	4.68	0.860
	HAGAS _{w/o} VLSTM	44.50	6.52	5.06	0.874
	HAGAS	49.10	6.22	4.74	0.964

The drift-aware agent grouping consists of two basic operations: (1) global assignment, and (2) local assignment. In (1) *global assignment*, we assign each agent (region) to one of a total C global agent group based on the closeness of their trajectory embeddings. In particular, we cluster $\{\mu_t^k\}_{k=1}^K$ into a total of C clusters (e.g., K-means in our current implementation). Within each global group c , HAGAS maintains a set of local agent groups. Within each the global agent group, we perform the following local assignment which further splits and merges the local agent groups. In (2) *local assignment*, the key idea is to split and merge the groups based on the divergence and convergence of the agent behaviors. In terms of splitting, we perform the bisecting K -Means (K is set as two) clustering to split a local group when the intra-group divergence and group size exceed thresholds, while two local groups within the same global group merge if the divergence goes below a threshold.

5 Experimental Evaluation

• **Datasets Studied.** Our experimental studies are based on the following two micromobility datasets, one on Citi Bike Sharing from NYC and one on e-scooter sharing from Austin, Texas. Specifically, we use a total of 996,144 trips recorded in January 2024, and aggregated them according to the census tracts. Our episodes have simulated a total of $T=31$ days of the study period. Regarding the e-scooter dataset from Austin, we similarly map the e-scooter trips in October 2021 to $K=127$ census tracts, which cover the downtown and university neighborhoods. Our episodes have simulated a total of $T=31$ days of the study period.

• **Baselines Studied.** We compare HAGAS against parameter-sharing extremes and recent dynamic sharing methods, including no-share (independent PPO), share-all (shared PPO), static grouping, dynamic sharing [7], and Kaleidoscope [5]. We also evaluate HAGAS without identity (ID) embeddings, without split-merge (SM) operations, without hierarchical grouping (HG), without adaptive grouping period (ARP), without trajectory encoder (VLSTM).

• **Overall Performance.** We evaluate performance based on Mobility Resource Availability $MRA = 1 - \frac{\sum_t \text{Unmet}_t}{\sum_t \text{Demand}_t}$, Moves Per Vehicle $MPV = \frac{\sum_t \text{Moves}_t}{N_{\text{Vehicle}}}$, Move Distances Per Vehicle $MDPV =$

$\frac{\sum_t \text{Dist}_t}{N_{\text{Vehicle}}}$, and Availability-Distance Score A-D Score $= a \cdot \frac{MRA}{\max MRA} + (1 - a) \cdot \frac{\min MDPV}{MDPV}$. Table 1 summarizes the quantitative comparison across the two deployments in both City of Austin and NYC. We can see that independent PPO (No-Share) showcases the drawback of over-parameterization without sharing the model parameters. We also show in Table 1 the percentages of relative differences of all the metrics studied. HAGAS achieves the highest MRA on both Austin and NYC, reaching 52.91% and 49.10%, respectively. Compared with the strongest baseline in each city, Dynamic Sharing in Austin and Independent PPO in NYC, HAGAS improves MRA by 14.8% and 8.0%, respectively. The movement-related metrics, MPV and MDPV, characterize the operational efforts required by each method.

• **Ablation Studies.** The ablation studies in Table 1 further verify the relative importance of each component of HAGAS. Removing the identity embeddings (w/o ID) causes the largest availability degradation, showing that agent-specific conditioning is important for preserving local regional behaviors under shared policies. Disabling split-merge (w/o SM), hierarchical grouping (w/o HG), the adaptive regrouping period (w/o ARP), or the VLSTM trajectory encoder (w/o VLSTM) also consistently reduces availability, which indicates that the adaptive grouping mechanism depends on both temporal trajectory representation and flexible group evolution. Overall, the ablation results support that the full HAGAS design improves service availability by combining global parameter sharing with local specialization and drift-aware regrouping.

6 Conclusion & Acknowledgment

We have designed and implemented HAGAS, a multi-agent reinforcement learning approach with hierarchical and adaptive grouping-based parameter sharing for dynamic mobility resource allocation. Using the bike sharing in NYC and e-scooter sharing in Austin, Texas as case studies, HAGAS addresses two challenges regarding MARL for mobility resource allocation — that is, (a) how to achieve global-local adaptivity to spatio-temporal demand-supply drifts, and (b) how to achieve global-local scalability in heterogeneous urban mobility resource allocation.

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